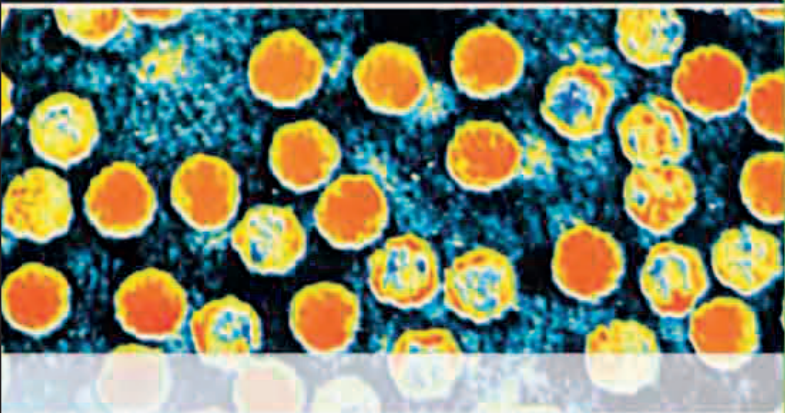




Life in a flask

Stanley Miller and Harold Urey recreated the dawn of life in a flask: their "primordial soup" produced amino acids within two weeks.



found to contain amino acids—the building blocks of proteins. This showed that the building blocks of life could be made from the simplest of substances.

In November, **Jonas Salk**, an American virologist, announced a breakthrough of a more humanitarian nature. He developed a **vaccine against poliomyelitis** based on a dead form of the polio virus. It was a far safer version of a live vaccine

that had previously been tried and found unsuccessful earlier.

Just as one discovery was being championed, another was being demolished. A **hominoid skull** in the Natural History Museum, London, supposedly excavated in Piltdown, East Sussex, in 1912 and acclaimed as a valuable link in the story of human evolution, was **exposed as a hoax**. The anatomists and paleontologists Kenneth Oakley, Wilfred Le Gros Clark, and Joseph Weiner declared that the skull actually consisted of a medieval human cranium, the teeth of a fossil chimpanzee, and the jaw of an orangutan. To this day, no one knows who perpetrated the hoax and duped the academic world for so long.

ROSALIND FRANKLIN (1920–58)

Trained in chemistry, Rosalind Franklin applied her skills to study the structure of biological molecules. She produced an X-ray diffraction image of DNA—"Photograph 51"—that showed a cross pattern. This suggested that DNA was helical in shape. It became a key piece of evidence in Watson and Crick's double helix model. Franklin died in 1958, and she did not share the Nobel Prize awarded for this achievement.



The Boeing 367-80—the prototype for the Boeing 707, which was the most successful of the early passenger jetliners—was donated to the Smithsonian Air and Space Museum in Washington, D.C. after being used to test design innovations.

ENCOURAGED BY EARLY RESULTS,

the US conducted a nationwide field trial—the largest medical field trial ever—of Salk's **polio vaccine**. On February 23, a **mass vaccination** program involving 1.8 million schoolchildren began. In 1955, a license was issued for its routine use. Salk's vaccine went on to protect children from polio across the world and heralded the World Health Organization's international campaign to eradicate polio.

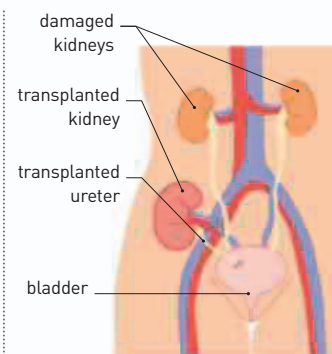
In May, the American aerospace company **Boeing** rolled out a new type of jet aircraft. The **367-80** was the prototype for the 707 passenger aircraft that came into use in the 1960s and 1970s. Until then, civil aviation had mostly been centered on aircraft that were propeller-driven, but the 367-80 demonstrated that **jet propulsion** was the way forward.

From 1954, scientists worldwide were able to apply a standard unit of measurement for temperature, following the General Conference on Weight and Measures held in France. The conference had been established to oversee what would be known as the **International System of Units (SI)**. In 1954, "kelvin" (named after British physicist Lord Kelvin) was deemed to be the SI unit for temperature.

At the same time, **Nikolay Basov** and **Alexander Prokhorov**

273.16 K THE TEMPERATURE AT WHICH WATER, ICE, AND VAPOR CAN COEXIST

at the USSR Academy of Sciences published their description of **maser** (microwave amplification by stimulated emission of radiation)—a system for concentrating beams of radiation. Maser came to be used in atomic clocks and helped amplify tiny signals in long-distance television broadcasts. Since then, researchers have explored its

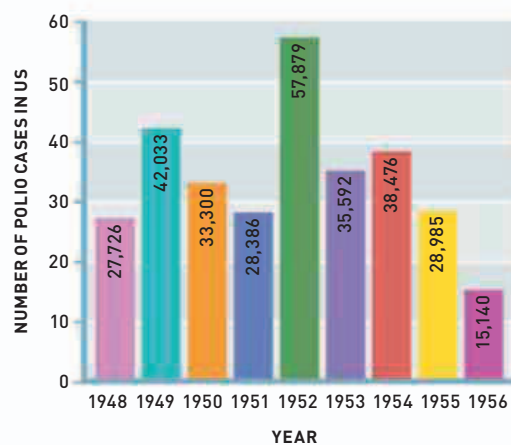


Kidney transplant

In a kidney transplant, the failed kidney is usually left in place. The donated kidney is implanted lower down the body and connected to a different part of the blood system.

potential for other uses, such as in medical body scanners.

The year ended with the first successful **kidney transplant**, carried out by **Joseph Murray** in Boston, Massachusetts.



Polio vaccinations

The use of Salk's polio vaccine for mass vaccination of children across the US brought down the number of polio cases by the thousands.

November 13
New York Times reports that Jonas Salk guarantees his antipolio vaccine

November 30
Time magazine reports that the Piltdown Man skull is a hoax

February 23
First field trial of antipolio vaccine starts in Virginia

March 1 The US carries out a hydrogen bomb test on Bikini Atoll

June 4 First description of how isolated chloroplasts carry out photosynthesis

May 14 Boeing unveils 367-80, the prototype for the Boeing 707

October 5-14 General Conference on Weight and Measures declares that the Kelvin is the standard unit of temperature

October 31 Basov-Prokhorov paper describes the principles of maser

November 30 First documented case of a meteorite hitting a human, in Sylacauga, Alabama

December 23
First successful organ transplant—of a kidney